94 days.

Average building approval time in Malaysia. Not because the building failed. Because the paperwork did.

The building was done. The documents were complete.
One street name.

ONBOARD

POWERED BY Z.AI GLM · REGULATED ONBOARDING ENGINE

02 / THE COST OF ONE WRONG LINE

21 business days to resubmit.

One address format difference. No missing documents. No failed inspection. Nobody built a system that could have caught it in seconds.

7+	3–4×	0
OFFICES PER SUBMISSION	AVERAGE RESUBMIT CYCLES	CENTRAL DIRECTORY

MBSA · Submission #BLD-2025-0419

SHAH ALAM — ZONING CLEARANCE REVIEW

REJECTED: Address mismatch on submitted certificate.

FOUND Jalan Ampang, KL
EXPECTED Jalan Imbi, KL

Next slot: 21 business days

⚠ Resubmit

Two streets apart. One month lost.

03 / WHO CARRIES THIS

Three people. Everyone's stuck.

Nobody here is failing. The admin is thorough. The owner is diligent. The problem is the absence of a system.

CA

01 — CITY ADMIN

The enforcer

"I know what we require."

No system to enforce it consistently. Every submission looks different. Manual review doesn't scale.

BO

02 — BUILDING OWNER

The applicant

"I submitted everything they asked for."

Three weeks later: one wrong address. Resubmit cycle two. Construction crew idle. Clock running.

PB

03 — THE PUBLIC

The tenant

"The building is done. I can see it."

But approved? Available? Nobody's built a place to check.

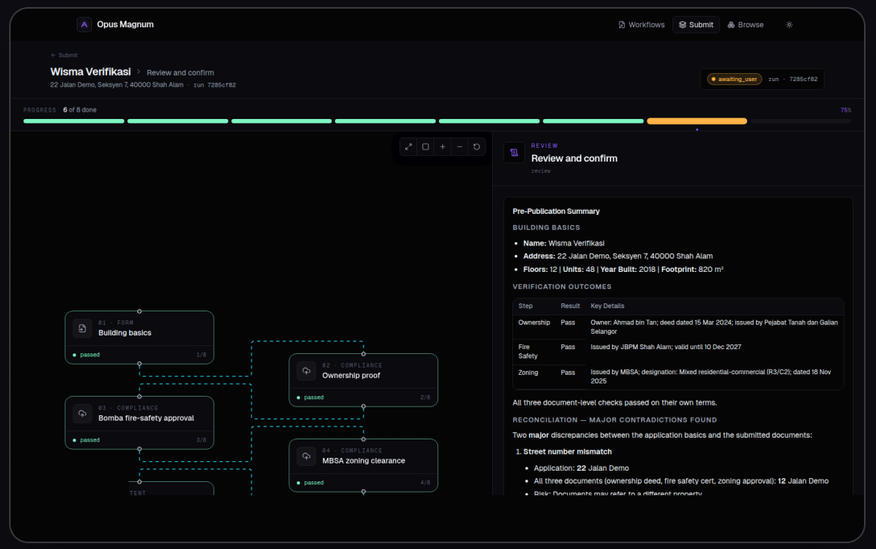
From blank form to published building. One sitting.

One platform. Three roles. Z.AI's GLM reads every document, verifies every claim, and publishes the result.

Admin Templates → Owner Uploads → GLM Verifies → Published → Visitor Explores

CORE INSIGHT

The workflow is data, not code. A new city is a new template. The same engine runs every one.



Onboarding canvas — every step run by GLM, publish summary on the right.

04B / WORKFLOW STRUCTURE

Six primitives.
Any city's process.

Every onboarding workflow is assembled from the same six building blocks.

📝

Form

Collects structured input — name, address, floor count.

🔐

Compliance Upload

GLM reads and verifies against admin-set criteria.

📁

Content Upload

Floor plans, photos — powers the visitor page.

🔄

Cross-Check

GLM reconciles all submissions against each other.

📋

Review

AI-generated summary. Owner confirms before publish.

🚀

Publish

Building goes live. Visitor page activates.

ADMIN TYPES →

"Shah Alam requires MBSA zoning clearance, Bomba fire-safety approval (Sijil Perakuan Bomba), proof of ownership, and standard floor plans + photos. Include a cross-check and a final review."

↓

GLM GENERATES →

01 **Basic Information Form**
building name, address, owner details

02 **Ownership Certificate**
compliance · verified against criteria

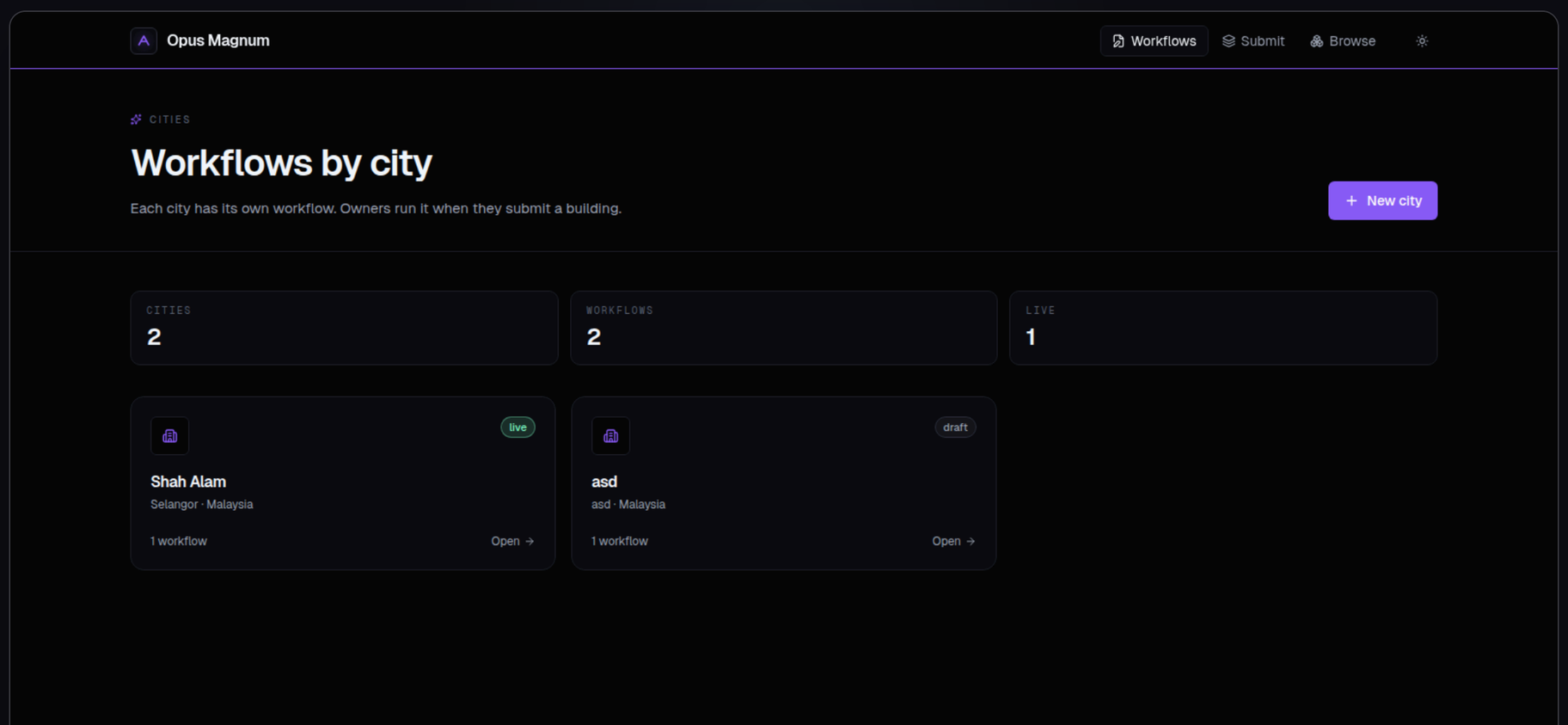
03 **MBSA Zoning Clearance**
compliance · address + zone match

04 **Bomba Fire-Safety (SPB)**
compliance · certificate + expiry

05 **Floor Plans + Photos**
content · powers visitor page

06 **Cross-Check**
reconcile all submitted data

07 **Review + Publish**
AI summary → owner confirms



05B / GLM CAPABILITIES

Five things only GLM makes possible.

Remove any one of these and the platform loses that function entirely. There is no fallback.

GLM_01

Template authoring

Plain-English description → complete workflow. Admins design processes without writing a single rule.

GLM_02

Document parsing

Reads PDFs, scans, images. Extracts owner name, address, certificate number from unstructured files.

GLM_03

Compliance verification

Applies admin criteria to document content. Returns a verdict with specific evidence — not just pass or fail.

GLM_04

Cross-document reconciliation

Reads all submissions together. Catches contradictions no single step would find alone.

GLM_05

Visitor assistant

The public page's chat reads the same files the workflow verified. One knowledge base. No separate pipeline.

GLM

Z.AI General Language Model

CENTRAL REASONING ENGINE

Remove GLM.
The system stops.

Not degrades. These five functions do not exist without GLM. There is no version that does less — only a version that does nothing.

✗ **No GLM** → No template drafting. Admins face a blank editor. Shah Alam's workflow doesn't exist.

✗ **No GLM** → No document reading. Uploaded files are binary blobs. Nothing is extracted. Nothing is known.

✗ **No GLM** → No compliance check. Forms submit. Nothing ever verifiably passes or fails.

✗ **No GLM** → No cross-check. The form says 10 floors. The plan shows 12. It publishes wrong.

✗ **No GLM** → No visitor assistant. The chat panel loads. There's nothing for it to say.

> The forms render. The buttons work. The platform does nothing.

07 / HOW GLM HANDLES REAL-WORLD FAILURES

Case 01 of 03 — GLM detects: Address Mismatch

📄 WHAT WAS SUBMITTED

📄 **Bomba Fire-Safety Certificate**
Sijil Perakuan Bomba · Compliance Upload · Step 4

A legitimate, valid certificate — properly signed and dated. GLM parses it successfully and extracts all fields.

CERTIFICATE SAYS	≠	REGISTERED ADDRESS
Jalan Ampang, KL		Jalan Imbi, KL

↩ Owner re-uploads the correct certificate

🔄 Step retries automatically — no restart needed

✓ Workflow continues from this point

💬 WHAT Z.AI GLM RETURNED

> *"This certificate reads Jalan Ampang. Your registered address is Jalan Imbi. These must match before this step can pass."*

GLM_03 · Compliance Verification · Step 4

⚡ WHY GLM WAS NEEDED HERE

A human reviewer checks one document at a time, in a queue. GLM reads the certificate and the registered address simultaneously, compares the two, and returns the specific mismatch — in seconds, not weeks.

Case 02 of 03 — GLM detects: Cross-Document Contradiction

WHAT WAS SUBMITTED

Two separate documents, two steps apart
Basics Form (Step 1) + Architectural Floor Plan (Step 5)

Neither document is wrong on its own. The contradiction only becomes visible when you read both together.

FORM SAYS	≠	FLOOR PLAN SHOWS
10 floors		12 floors

- ✏️ Owner corrects the form to match the floor plan
- 🔄 Cross-check step retries automatically
- ✓ Accurate record published

WHAT Z.AI GLM RETURNED

> *"The floor plan shows 12 floors. Your basics form states 10. Reconcile before this step can pass."*

GLM_04 · Cross-Document Reconciliation · Step 6

⚡ WHY GLM WAS NEEDED HERE

This contradiction lives across two documents uploaded at different steps. No single-step verification would catch it. GLM reads all submitted documents simultaneously at the cross-check step — a task no sequential human review handles reliably.

Case 03 of 03 — GLM detects: Missing Required Field

🔍 WHAT WAS SUBMITTED

🏛️ **MBSA Zoning Clearance Certificate**
Compliance Upload · Step 3

The certificate is legitimate and was parsed successfully. All visible fields read correctly — except one is simply absent from the document.

MISSING FIELD
Certificate Expiry Date
Required by Shah Alam workflow criteria

📨 Owner requests reissued certificate from MBSA
🔄 Re-uploads — step retries from this point
✓ Workflow resumes — no earlier steps repeated

💬 WHAT Z.AI GLM RETURNED

> *"The certificate was parsed successfully. The expiry date field is absent. This field is required by Shah Alam criteria."*

GLM_03 · Missing Field Detection · Step 3

⚡ WHY GLM WAS NEEDED HERE

The document looks valid on the surface. GLM parses it, maps every field against the admin's criteria, and identifies the gap — not a format error, but an absent field. The system doesn't fail silently. It pauses, explains, and tells the owner exactly what to fix.

✓ Ambiguous input handling
✓ Plain-English rejections
✓ Step-level retry
✓ Stateful resumption
✓ Incomplete data detection

We built this for buildings.

The engine adapts. The AI stays. Same platform, same primitives, different template.

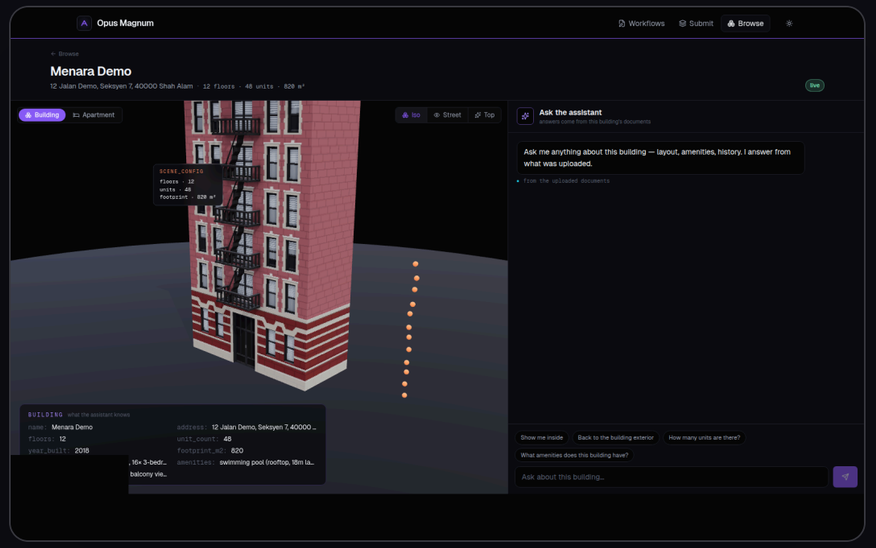
Building Onboarding
Shah Alam. MBSA. Live MVP.

▶ LIVE MVP

HR Onboarding
Same engine. New template.

KYC & Compliance
Regulated, multi-document.

Vendor Licensing
Permits, procurement.



Published building — 3D scene, profile readout, grounded assistant chat.

> "From blank form to published building — *guided, verified, and published by Z.AI's GLM.*
>
> *You already know what comes next."*

ONBOARD · UMHACKATHON 2026 · DOMAIN 1